

Series Expansions for the Solution of an Integral-Functional Equation with a Parameter

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Abstract. We continue our considerations in [2] and [3] on a homogeneous integral-functional equation with a parameter $a > 1$ and derive different series expansions for the solution which simplify in the case $a \geq 2$. The terms of these series can be interpreted as polynomial splines.

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1. Introduction

There exists a long history concerning compactly supported C^∞ -functions, which are solutions of differential-functional equations, cf. Rvachev [8] and the literature quoted there. By integration these equations can be transformed into integral-functional equations. A special example of such an equation reads

$$\phi(t) = b \int_{at-a+1}^{at} \phi(\tau) d\tau \quad \left(b = \frac{a}{a-1} \right) \quad (1.1)$$

with the real variable t and a positive parameter $a \neq 1$. The solutions of (1.1) were studied for $a = 3$ in Wirsching [13], for $a > \frac{3}{2}$ in Wirsching [14], and for $a = 2$ in Schnabl [10] and Volk [12]. For $a > 1$ it was shown in [2] that (1.1) has a C^∞ -solution with the support $[0, 1]$, which is uniquely determined after the normalization

$$\int_0^1 \phi(t) dt = 1. \quad (1.2)$$

For $0 < a < 1$ equation (1.1) has only the zero solution.

Jia, Lee and Sharma [5] have studied the spectral properties of the integral equation

$$\phi(t) = a \int_{-\infty}^{+\infty} c(s) \phi(at - s) ds \quad (1.3)$$

with $a > 1$ which generalizes the case $a = 2$ in Derfel, Dyn and Levin [4]. For $c = \frac{1}{a-1} \chi_{[0, a-1]}$ equation (1.3) turns over into (1.1). The (analogously to (1.2) normalized) solution of (1.3) in the case $c = \frac{1}{2(a-1)} \chi_{[-a+1, a-1]}$ is the generalized up-function, which is connected with the solution of (1.1)-(1.2) by $\phi(t) = 2 \operatorname{up}(2t - 1)$. The generalized up-function was studied by Kabaya and Iri in [6] and [7] with the notations p_∞ and $a = \frac{1}{1-\alpha}$. Originally, the up-function was considered only in the case $a = 2$, cf. Rvachev and Rvachev [9] and the detailed surveys in Rvachev [8] and Sharma [11].

In the following we derive new series expansions for the solution of (1.1)-(1.2) by means of polynomial splines. We point out that such series also exist for the solutions of related integral equations. Partly, the series are based on special series expansions for real numbers and on nested interval representations with open intervals.

Preliminary, we list those results from [2] and [3] which are needed later on. The solution of (1.1)-(1.2) with $a > 1$ is symmetric with respect to the point $\frac{1}{2}$, monotone at both sides of $\frac{1}{2}$, and it is strictly positive for $t \in (0, 1)$ (whereas $\phi(0) = \phi(1) = 0$). The solution ϕ defines Appell polynomials ψ_n by means of the representations

$$\psi_n(t) = \int_0^1 (t - \tau)^n \phi(\tau) d\tau \quad (n \in \mathbb{N}_0). \quad (1.4)$$

In the case $a \geq 2$, these equations can partly be inverted, namely e.g.

$$\phi(t) = c_n \psi_n(a^{n+1}t) \quad (1.5)$$

for $\frac{1}{a^{n+1}} \leq t \leq \frac{a-1}{a^{n+1}}$ where c_n is given by

$$c_n = \frac{b^{n+1}}{n! a^{\frac{n(n+1)}{2}}} = \frac{1}{n! a^{\frac{(n+1)(n-2)}{2}} (a-1)^{n+1}}. \quad (1.6)$$

The polynomials ψ_n can also be calculated recursively by $\psi_0(t) = 1$ and

$$\psi_n(t) = \left(t - \frac{1}{2}\right) \psi_{n-1}(t) + \frac{1}{n} \sum_{\nu=2}^n \binom{n}{\nu} B_\nu \frac{(a-1)^\nu}{a^\nu - 1} \psi_{n-\nu}(t) \quad (n \in \mathbb{N}), \quad (1.7)$$

where B_ν are the Bernoulli numbers in the notation of [1]. Note that in [2] we have used the abbreviation ψ_n for the polynomials (1.4) with $\frac{1}{a}$ instead of a in the solution ϕ . The polynomials (1.4) have the properties

$$\psi'_n(t) = n \psi_{n-1}(t), \quad (1.8)$$

$$\psi_n(1-t) = (-1)^n \psi_n(t), \quad (1.9)$$

$$\psi_n(t) - \psi_n(t - a + 1) = na^{n-1}(a-1)\psi_{n-1}\left(\frac{t}{a}\right), \quad (1.10)$$

($n \in \mathbb{N}_0$), and the special values $\psi_n(1)$ satisfy the inequalities

$$\frac{1}{2^n} \leq \psi_n(1) \leq \frac{1}{n+1}. \quad (1.11)$$

For $n \geq 1$ and $t > \frac{1}{2}$, the polynomials $\psi_n(t)$ are positive and strongly increasing.

We recall the auxiliary sequence $\gamma_n = \gamma_n(a)$ ($n \in \mathbb{N}_0$) defined as follows: If n has the dyadic representation $n = d_m \dots d_1 d_0$ with $d_\nu \in \{0, 1\}$ then

$$\gamma_n = (a-1) \sum_{\nu=0}^m d_\nu a^\nu. \quad (1.12)$$

The sequence γ_n can also be defined recursively by $\gamma_0 = 0$, $\gamma_1 = a-1$ and

$$\gamma_{2k} = a\gamma_k, \quad \gamma_{2k+1} = a\gamma_k + \gamma_1 \quad (k \in \mathbb{N}_0), \quad (1.13)$$

in particular $\gamma_n = n$ for $a = 2$. In the case $a > 2$, the sequence γ_n satisfies

$$\gamma_n + 1 < \gamma_{n+1}, \quad (1.14)$$

and defines the following Cantor intervals

$$G_{kn} = \left(\frac{\gamma_{2k} + 1}{a^{n+1}}, \frac{\gamma_{2k+1}}{a^{n+1}} \right), \quad F_{kn} = \left[\frac{\gamma_k}{a^n}, \frac{\gamma_k + 1}{a^n} \right] \quad (1.15)$$

$k \in \{0, 1, \dots, 2^n - 1\}$ ($n \in \mathbb{N}_0$) with $G_{kn} \subset F_{kn}$.

Moreover, we recall the sign sequence $\varepsilon_k = (-1)^{\nu(k)}$, where $\nu(k)$ denotes the number of "1s" in the dyadic representation of n , i.e. $\nu(k)$ is the binary sum-of-digits function. The numbers ε_k satisfy the equalities

$$\sum_{k=0}^{2^n-1} \varepsilon_k = 0 \quad (n \in \mathbb{N}). \quad (1.16)$$

The functions ϕ_n ($n \in \mathbb{N}_0$) are defined recursively by $\phi_0 = \phi$ from (1.1)-(1.2) and

$$\phi_{n+1}(t) = \int_0^t \phi_n(\tau) d\tau, \quad (1.17)$$

they satisfy the integral equations

$$\phi_n(t) = \frac{b}{a^n} \int_{at-a+1}^{at} \phi_n(\tau) d\tau \quad \left(b = \frac{a}{a-1} \right) \quad (1.18)$$

and for arbitrary $a > 1$ and $n \in \mathbb{N}$

$$\phi_n(t) = \frac{1}{(n-1)!} \psi_{n-1}(t) \quad (t \geq 1). \quad (1.19)$$

For $a > 1$, $t \in \mathbb{R}$ and $n \in \mathbb{N}_0$, the solution of (1.1)-(1.2) possesses the representations

$$\phi(t) = \frac{b^n}{a^{\frac{n(n-1)}{2}}} \sum_{k=0}^{2^n-1} \varepsilon_k \phi_n(a^n t - \gamma_k). \quad (1.20)$$

Unfortunately, in the case $a = 2$, the special values

$$\begin{aligned} \phi\left(\frac{1}{2}\right) &= 2, \quad \phi\left(\frac{1}{4}\right) = 1, \quad \phi\left(\frac{1}{8}\right) = \frac{5}{36}, \quad \phi\left(\frac{3}{8}\right) = \frac{67}{36}, \\ \phi\left(\frac{1}{16}\right) &= \frac{1}{144}, \quad \phi\left(\frac{3}{16}\right) = \frac{73}{144}, \quad \phi\left(\frac{5}{16}\right) = \frac{215}{144}, \quad \phi\left(\frac{7}{16}\right) = \frac{287}{144} \end{aligned}$$

were incorrect in [3], disregarding the first two.

2. Auxiliary inequalities

We begin with some inequalities which will play a basic role later on. For given a and t our emphasis lies on all pairs $(k, n) \in \mathbb{N}_0 \times \mathbb{N}_0$ with

$$k < 2^n \quad (2.1)$$

such that

$$\frac{1}{a} \leq a^n t - \gamma_k < 1 \quad (2.2)$$

with γ_k defined by (1.12). For positive a and real t the inequality (2.2) can have solutions with (2.1) only in the case that

$$a > 1 \quad \text{and} \quad 0 < t < 1. \quad (2.3)$$

Lemma 2.1 *For a fixed $k \in \mathbb{N}_0$ there exists at most one $n \in \mathbb{N}_0$ satisfying (2.2).*

Proof. Assume that for fixed $k \in \mathbb{N}_0$ there are two $n, m \in \mathbb{N}_0$ satisfying (2.2) and $\frac{1}{a} \leq a^m t - \gamma_k < 1$, respectively. The last inequalities are equivalent to

$$-a^{n-m} < a^{n-m} \gamma_k - a^n t \leq -a^{n-m-1}$$

and by addition to (2.2) we get the inequalities

$$\frac{1}{a} - a^{n-m} < (a^{n-m} - 1) \gamma_k < 1 - a^{n-m-1}$$

which imply $m = n$ in view of $\gamma_k \geq 0$ and $a > 1$ ■

According to (2.3) the first inequality in (2.2) is satisfied for $n = n_k$ where

$$n_k = \left\lceil \frac{\ln(\gamma_k + \frac{1}{a}) - \ln t}{\ln a} \right\rceil, \quad (2.4)$$

and where $\lceil \cdot \rceil$ denotes the ceiling function, cf. [13], p. 52.

Definition 2.2 For given a, t with (2.3) we define a sequence $\delta_k = \delta_k(a, t)$ ($k \in \mathbb{N}_0$) by $\delta_k = 1$ if both (2.1) and (2.2) are satisfied for $n = n_k$, and by $\delta_k = 0$ elsewhere.

This means in connection with Lemma 2.1 that there are no other solutions (k, n) of (2.1)-(2.2) except those with $\delta_k = 1$ and $n = n_k$. Note that $\gamma_k < a^{n_k}$ and therefore $k < 2^{n_k}$ for $a \geq 2$, but in the case $a < 2$ the inequality (2.1) can be violated for $n = n_k$.

The inequality (2.2) can be written as

$$\frac{\gamma_k}{a^n} + \frac{1}{a^{n+1}} \leq t < \frac{\gamma_k + 1}{a^n} \quad (2.5)$$

and shall be compared with

$$\frac{\gamma_\ell}{a^m} < t < \frac{\gamma_\ell + 1}{a^m} \quad (m, \ell \in \mathbb{N}_0). \quad (2.6)$$

Lemma 2.3 The inequalities (2.2) and (2.6) are equivalent in the following sense: If (2.2) is satisfied then also (2.6) with $\ell = k$ and $m = n$. If (2.6) is satisfied, then there exists a uniquely determined $j \in \mathbb{N}_0$ such that (2.2) is satisfied with $n = m + j$ and $k = 2^j \ell$.

Proof. The first assertion is trivial. The second follows from the disjoint decomposition

$$\left(\frac{\gamma_\ell}{a^m}, \frac{\gamma_\ell + 1}{a^m} \right) = \bigcup_{j=0}^{\infty} \left[\frac{\gamma_{2^j \ell} + 1}{a^{m+j+1}}, \frac{\gamma_{2^j \ell} + 1}{a^{m+j}} \right),$$

cf. the first equality in (1.13) ■

Remark 2.4 In the case $1 < a < 2$ the inequality (2.2) with fixed $n \in \mathbb{N}$ can have several solutions k with the property (2.1), where the number of these k is unbounded for $a \rightarrow 1+0$. In particular, it can happen that two γ_k are equal for different k . Examples for the last assertion follow from the equality

$$\gamma_{2^j m} - \gamma_{2^j m-1} = a^{j+1} - 2a^j + 1, \quad (2.7)$$

which holds for odd $m \in \mathbb{N}$ and $j \in \mathbb{N}$. We only have to choose $a = w_j$ ($j \geq 2$) the uniquely determined zero of $w^{j+1} - 2w^j + 1$ with $w_j > 1$. By means of the formula of Lagrange (cf. [1]) one finds the convergent expansion

$$w_j = 2 - \sum_{\nu=1}^{\infty} \frac{1}{\nu} \binom{\nu j + \nu - 2}{\nu - 1} 2^{1-(j+1)\nu}$$

which is simultaneously an asymptotic expansion as $j \rightarrow \infty$, so that e.g.

$$w_j = 2 - \frac{1}{2^j} - \frac{j}{2^{2j+1}} + \mathcal{O}\left(\frac{j^2}{2^{3j}}\right).$$

Lemma 2.5 *In the case $a \geq 2$ and for fixed $n \in \mathbb{N}_0$ there exists at most one $k \in \mathbb{N}_0$ such that (2.2) is satisfied.*

Proof. Assume for fixed $n \in \mathbb{N}_0$ that there are two $k, \ell \in \mathbb{N}_0$ satisfying (2.2) and $\frac{1}{a} \leq a^n t - \gamma_\ell < 1$, respectively. The last inequalities are equivalent to

$$-1 < \gamma_\ell - a^n t \leq -\frac{1}{a}$$

and by addition to (2.2) we obtain the inequalities

$$-1 + \frac{1}{a} < \gamma_\ell - \gamma_k < 1 - \frac{1}{a}, \quad (2.8)$$

which imply $k = \ell$ in view of (1.14) for $a > 2$, and $\gamma_k = k$ for $a = 2$ ■

In the following we use the abbreviation

$$f_n(t) = c_n \psi_n(t) \quad (2.9)$$

with c_n from (1.6) and ψ_n from (1.7) so that (1.5) simplifies to

$$\phi(t) = f_{n-1}(a^n t) \quad (2.10)$$

for $\frac{1}{a^n} \leq t \leq \frac{\gamma_1}{a^n}$ ($n \in \mathbb{N}$), and (1.10) to

$$f_n(t) - f_n(t - \gamma_1) = f_{n-1}\left(\frac{t}{a}\right). \quad (2.11)$$

Lemma 2.6 *In the case $a \geq 2$ there are valid the inequalities*

$$f_n(t) > f_{n+m}(a^m(t - \gamma_1)), \quad (2.12)$$

for $(m, n) \in \mathbb{N} \times \mathbb{N}_0 \setminus \{(1, 0)\}$ and

$$\gamma_1 \left(1 + \frac{1}{a^m}\right) \leq t < \gamma_1 + \frac{1}{a^{m-1}}. \quad (2.13)$$

Proof. Using (1.4), (1.6) and (2.9), the lemma is proved if

$$\frac{1}{n!} (n+m)! (a-1)^m a^{\frac{m(2n+m-1)}{2}} (t-\tau)^n > (a^m(t-\gamma_1) - \tau)^{n+m} \quad (2.14)$$

for $a \geq 2$, t from (2.13) and $0 \leq \tau \leq 1$. In view of $a^m(t - \gamma_1) - \tau \geq 0$ for these t, τ , the inequality (2.14) holds if both

$$a^m(t - \tau) > a^m(t - \gamma_1) - \tau$$

and

$$\frac{1}{n!} (n+m)! (a-1)^m a^{\frac{m(m-1)}{2}} > (a^m(t - \gamma_1) - \tau)^m.$$

The first inequality is equivalent to $a^m(a-1) > (a^m-1)\tau$ and this is valid in view of $\tau \leq 1$. The second inequality is valid if

$$\frac{1}{n!}(n+m)!(a-1)^m a^{\frac{m(m-1)}{2}} \geq a^m.$$

In the case $m = 1$ this means $(n+1)(a-1) \geq a$ and it is satisfied for $n \geq 1$. In the case $m = 2$ this means $(n+1)(n+2)(a-1)^2 \geq a$ and it is satisfied for $n \geq 0$, and in the case $m \geq 3$ the second inequality is also satisfied for $n \geq 0$ in view of $(a-1)a^{\frac{m-3}{2}} \geq 1$, all inequalities subject to $a \geq 2$ ■

Remark 2.7 For the validity of the inequality (2.12) the interval (2.13) is not optimal, but it suffices for a later application.

3. The main expansion theorems

Now, we come to our main results where we recall besides (1.12) and (2.9) the definitions of ε_k before (1.16), n_k in (2.4) and δ_k in Definition 2.2.

Theorem 3.1 *For each $a > 1$ and $t \in (0, 1)$ the solution ϕ of (1.1)-(1.2) has the expansion*

$$\phi(t) = \sum_{k=0}^{\infty} \varepsilon_k \delta_k f_{n_k}(a^{n_k+1}t - \gamma_{2k}), \quad (3.1)$$

which converges uniformly in $(0, 1)$.

Proof. In view of (1.17) equation (1.18) with $t = \tau$ and $n = m$ can be written in the form

$$\phi_m(\tau) = \frac{b}{a^m} [\phi_{m+1}(a\tau) - \phi_{m+1}(a\tau - \gamma_1)] \quad (m \in \mathbb{N}_0). \quad (3.2)$$

Substituting $\tau = a^m t - \gamma_k$ ($m, k \in \mathbb{N}_0$) and using (1.13) we obtain

$$\phi_m(a^m t - \gamma_k) = \frac{b}{a^m} [\phi_{m+1}(a^{m+1}t - \gamma_{2k}) - \phi_{m+1}(a^{m+1}t - \gamma_{2k+1})]. \quad (3.3)$$

If we apply this formula successively for $m = 0, 1, \dots, n-1$ and $k \in \{0, 1, \dots, 2^m-1\}$, then we obtain the already known representation (1.20). In order to prove (3.1), we apply (3.3) only in cases m, k with $a^m t - \gamma_k < 1$, so in particular for $m = k = 0$. According to

$$\left. \begin{aligned} a^{m+1}t - \gamma_{2k} &= a(a^m t - \gamma_k), \\ a^{m+1}t - \gamma_{2k+1} &= a(a^m t - \gamma_k - 1) + 1 \end{aligned} \right\} \quad (3.4)$$

the assumption $a^m t - \gamma_k < 1$ always implies $a^{m+1}t - \gamma_{2k+1} < 1$, whereas $a^{m+1}t - \gamma_{2k} \geq 1$ if and only if $\delta_k = 1$ and $m = n_k$. In the last case we do not apply (3.3) any longer, but we use (1.19) and (2.9), i.e. in view of $\varepsilon_{2k} = \varepsilon_k$ we use

$$\frac{b^{m+1}}{a^{\frac{m(m+1)}{2}}} \varepsilon_{2k} \phi_{m+1}(a^{m+1}t - \gamma_{2k}) = \varepsilon_k f_m(a^{m+1}t - \gamma_{2k}),$$

where the factor of ϕ_{m+1} was taken from (1.20) with $n = m + 1$. After n such steps we obtain for $\phi(t)$ the representation

$$\phi(t) = \sum_{k=0}^{2^{n-1}-1} \varepsilon_k \delta_k f_{n_k}(a^{n_k+1}t - \gamma_{2k}) + R_n \quad (3.5)$$

($n \in \mathbb{N}$) with the following meaning: The asterisk shall indicate that the sum contains only terms with $n_k < n$, and

$$R_n = \frac{b^n}{a^{\frac{n(n-1)}{2}}} \sum_{k \in A_n} \varepsilon_k \phi_n(a^n t - \gamma_k) \quad (3.6)$$

where A_n is the subset of numbers $k \in \{0, 1, \dots, 2^n - 1\}$ with $a^n t - \gamma_k < 1$, since $a^m t - \gamma_k \geq 1$ implies $a^{m+1} t - \gamma_\ell \geq 1$ both for $\ell = 2k$ and $\ell = 2k + 1$ according to (3.4).

For $n \in \mathbb{N}$ the functions $\phi_n(\cdot)$ are non-negative and increasing, so that in view of (1.11) and (1.19) we have

$$0 \leq \phi_n(\tau) \leq \phi_n(1) \leq \frac{1}{n!} \quad (3.7)$$

for $\tau < 1$. Moreover, since A_n contains at most 2^n elements and since $|\varepsilon_k| = 1$, we find the estimate

$$|R_n| \leq \frac{b^n}{a^{\frac{n(n-1)}{2}}} \sum_{k \in A_n} \phi_n(a^n t - \gamma_k) \leq \frac{(2b)^n}{n! a^{\frac{n(n-1)}{2}}} \quad (3.8)$$

so that $R_n \rightarrow 0$ as $n \rightarrow \infty$, and (3.5) converges to (3.1).

In order to show the uniform convergence, we write (3.1) in the form

$$\phi(t) = \sum_{k=0}^{K-1} \varepsilon_k \delta_k f_{n_k}(a^{n_k+1}t - \gamma_{2k}) + r_K$$

($K \in \mathbb{N}$) where

$$r_K = \sum_{k=K}^{\infty} \varepsilon_k \delta_k f_{n_k}(a^{n_k+1}t - \gamma_{2k}),$$

and note that $2^{n_k} > K$ for $k \geq K$. Now, we consider an arbitrary n with $2^n > K$. In view of (1.2) and $\phi(t) \geq 0$ the representation (1.4) implies $0 < \psi_n(\tau) < \tau^n$ for $1 \leq \tau$. Consequently, according to (1.6) and (2.9) we obtain

$$f_n(\tau) = c_n \psi_n(\tau) < c_n a^n = \frac{b^{n+1}}{n! a^{\frac{n(n-1)}{2}}} \quad (3.9)$$

for $1 \leq \tau < a$. According to (2.1) there are at most 2^n numbers k with $n_k = n$. Hence, we find in view of $|\delta_k| \leq 1$, $|\varepsilon_k| = 1$ and (2.2) for $n = n_k$ the estimate

$$|r_K| < \sum_{n=m}^{\infty} \frac{2^n b^{n+1}}{n! a^{\frac{n(n-1)}{2}}} \quad (3.10)$$

where $m = \lceil \frac{\ln(2K)}{\ln 2} \rceil$, which proves the uniform convergence of the series (3.1) ■

The expansion (3.1) can be interpreted as a series of polynomial splines.

Theorem 3.2 *For each $a > 1$ the solution ϕ of (1.1)-(1.2) has the expansion*

$$\phi(t) = \frac{1}{2} \sum_{n=0}^{\infty} \sum_{k=0}^{2^{n+1}-1} (-1)^k \varepsilon_k \chi_{kn} f_n(a^{n+1}t - \gamma_k) \quad (3.11)$$

where $\chi_{2k,n}$ is the characteristic function of the interval (2.5), and where χ_{kn} for odd k is the characteristic function of

$$\frac{\gamma_k - \gamma_1}{a^{n+1}} < t \leq \frac{\gamma_k}{a^{n+1}}. \quad (3.12)$$

The partial sums of (3.11) with respect to n have the property (1.2), and they are symmetric with respect to $\frac{1}{2}$.

Proof. Obviously, (3.1) can be written as $\phi(t) = \lim_{N \rightarrow \infty} S_N(t)$ where

$$S_N(t) = \sum_{n=0}^N \sum_{k=0}^{2^n-1} \varepsilon_k \chi_{2k,n} f_n(a^{n+1}t - \gamma_{2k}). \quad (3.13)$$

In view of (1.8), (2.9) and (2.11) for $t = a$ it holds

$$\int_0^1 \chi_{2k,n} f_n(a^{n+1}t - \gamma_{2k}) dt = \frac{1}{b} [f_{n+1}(a) - f_{n+1}(1)] = \frac{1}{b} f_n(1).$$

Hence, these integrals are independent of k , and according to $f_0 = b$ and (1.16), we find that all S_N satisfy (1.2). This also comes true for the linear combination

$$\frac{1}{2} [S_N(t) + S_N(1-t)], \quad (3.14)$$

which is symmetric with respect to $\frac{1}{2}$, and therefore also convergent to ϕ as $N \rightarrow \infty$. It remains to show that (3.14) are the partial sums of (3.11). For every even integer $2k$ with (2.1), we introduce its odd dyadic complement $k' = 2^{n+1} - 2k - 1$, for which it holds $\gamma_{k'} = a^{n+1} - 1 - \gamma_{2k}$ by means of (1.12) as well as $\nu(k') + \nu(2k) = n + 1$ and hence $\varepsilon_{k'} \varepsilon_k = (-1)^{n+1}$ according to $\nu(2k) = \nu(k)$. Consequently $\varepsilon_k = (-1)^{n+1} \varepsilon_{k'}$, and in view of (1.9) and (2.9) we obtain

$$\varepsilon_k f_n(a^{n+1}(1-t) - \gamma_{2k}) = -\varepsilon_{k'} f_n(a^{n+1}t - \gamma_{k'}).$$

Replacing t in (2.5) by $1-t$ this inequality turns over into (3.12) with k' instead of k , and we see that the sums (3.14) are indeed the partial sums of (3.11) ■

Supplement 3.3 Analogously as in Theorem 3.1 it can be proved for the solutions of (1.18) with $n \in \mathbb{N}$ that

$$\phi_n(t) = \frac{a^{\frac{n(n-1)}{2}}}{b^n} \sum_{k=0}^{\infty} \varepsilon_k \delta_k f_{n_k+n}(a^{n_k+1}t - \gamma_{2k}) \quad (3.15)$$

for $t \in (0, 1)$. The corresponding series for $\phi_{n-1} = \frac{d}{dt}\phi_n$ arises by termwise differentiation of (3.15) using the right derivative at the left endpoints of (2.5). Termwise integration of the series for ϕ_{n-1} yields (3.15) plus a series of step functions which must vanish identically for $t \in (0, 1)$. Repeated termwise differentiation of (3.1) yields corresponding series for $\phi^{(n)}$ ($n \in \mathbb{N}$), but termwise integration of the series for $\phi^{(n)}$ yields directly only the terms of the series for $\phi^{(n-1)}$ which are no step functions, the latter originate from the additional series of step functions.

4. Nested intervals and series of numbers

In the following it is useful to modify the intervals (1.15) by

$$\overline{G}_{kn} = \left[\frac{\gamma_{2k} + 1}{a^{n+1}}, \frac{\gamma_{2k+1}}{a^{n+1}} \right], \quad \mathring{F}_{kn} = \left(\frac{\gamma_k}{a^n}, \frac{\gamma_{k+1}}{a^n} \right), \quad (4.1)$$

$k \in \{0, 1, \dots, 2^n - 1\}$ ($n \in \mathbb{N}_0$), and to consider also the general case $a > 1$ where $\overline{G}_{kn}$ degenerates to one point $\frac{2k+1}{2^n}$ for $a = 2$, and $\overline{G}_{kn} = \emptyset$ for $a < 2$. Further, we introduce the set

$$M = \bigcup_{n=0}^{\infty} \bigcup_{k=0}^{2^n-1} \overline{G}_{kn}$$

and its complement $CM = (0, 1) \setminus M$ which can also be represented as

$$CM = \bigcap_{n=0}^{\infty} \bigcup_{k=0}^{2^n-1} \mathring{F}_{kn}. \quad (4.2)$$

Obviously, $M = \emptyset$ and $CM = (0, 1)$ for $a < 2$. Moreover,

$$\mathring{F}_{kn} = \mathring{F}_{2k,n+1} \cup \overline{G}_{kn} \cup \mathring{F}_{2k+1,n+1} \quad (4.3)$$

where the three sets on the right-hand side are disjoint in the case $a \geq 2$, whereas

$$\mathring{F}_{2k,n+1} \cap \mathring{F}_{2k+1,n+1} = \left(\frac{\gamma_{2k+1}}{a^{n+1}}, \frac{\gamma_{2k} + 1}{a^{n+1}} \right) \quad (4.4)$$

is non-empty in the case $a < 2$. The decomposition (4.3) shows that

$$\{\mathring{F}_{k_n,n}\} \quad (n \in \mathbb{N}_0, k_{n+1} \in \{2k_n, 2k_n + 1\}) \quad (4.5)$$

is a sequence of nested intervals which always begins with $k_0 = 0$. Since the intervals $\mathring{F}_{k_n}$ are open, it is possible that the intersection of the intervals (4.5) is empty. However, if it contains a point t , then t is uniquely determined by the nested intervals and we write $t = \{\mathring{F}_{k_n,n}\}$.

Note that only in the case $a \geq 2$ the sequence k_n is uniquely determined for a given $t \in (0, 1)$. In the case $1 < a < 2$ the decomposition (4.3) reduces to $\overset{\circ}{F}_{kn} = \overset{\circ}{F}_{2k,n+1} \cup \overset{\circ}{F}_{2k+1,n+1}$ in view of $\overline{G}_{kn} = \emptyset$, and for $t \in \overset{\circ}{F}_{k_n,n}$ we have

$$k_{n+1} = 2k_n \quad \text{if } t \in \overset{\circ}{F}_{2k_n,n+1} \setminus \overset{\circ}{F}_{2k_n+1,n+1} = \left(\frac{\gamma_{2k_n}}{a^{n+1}}, \frac{\gamma_{2k_n+1}}{a^{n+1}} \right], \quad (4.6)$$

$$k_{n+1} = 2k_n + 1 \quad \text{if } t \in \overset{\circ}{F}_{2k_n+1,n+1} \setminus \overset{\circ}{F}_{2k_n,n+1} = \left[\frac{\gamma_{2k_n} + 1}{a^{n+1}}, \frac{\gamma_{2k_n+1} + 1}{a^{n+1}} \right), \quad (4.7)$$

whereas k_{n+1} is not fixed if t is contained in (4.4) with $k = k_n$. But we can say more.

Lemma 4.1 *In the case $1 < a < 2$ there exist infinitely many possibilities to express a number $t \in (0, 1)$ by a sequence (4.5) of nested intervals.*

Proof. If $t \in \overset{\circ}{F}_{k_n,n}$ then, as we have just seen, only in the intervals appearing in (4.6) and (4.7) the index k_{n+1} of $\overset{\circ}{F}_{k_{n+1},n+1}$ is uniquely determined. More generally, only in the intervals

$$\left(\frac{\gamma_{2^j k}}{a^{n+j}}, \frac{\gamma_{2^j k+1}}{a^{n+j}} \right], \quad \left[\frac{\gamma_{2^j k} + 1}{a^{n+j}}, \frac{\gamma_{2^j k+1} + 1}{a^{n+j}} \right) \quad (4.8)$$

where $k = k_n$ and $j \in \mathbb{N}$ the index k_{n+j} of $\overset{\circ}{F}_{k_{n+j},n+j}$ is uniquely determined. Since

$$\frac{\gamma_{2^j k}}{a^{n+j}} = \frac{\gamma_k}{a^n}, \quad \frac{\gamma_{2^j k} + 1}{a^{n+j}} = \frac{\gamma_k + 1}{a^n},$$

are independent of j , and the lengths $\frac{\gamma_1}{a^{n+j}}$ of the intervals (4.8) are tending to zero as $j \rightarrow \infty$, there must exist a special j such that both cases for k_{n+j} in $\overset{\circ}{F}_{k_{n+j},n+j}$ are possible. Applying the same consideration with $n+j$ instead of n , we see by induction that Lemma 4.1 is true ■

The foregoing considerations yield

Lemma 4.2 *For $1 < a < 2$ the solutions (k, n) of (2.2) are exactly the pairs (k_n, n) , for which there is never $k_{n+1} = 2k_n$ in the representations $t \in \{\overset{\circ}{F}_{k_n,n}\}$.*

Remark 4.3 From (1.12) it follows that always $\gamma_{2m+1} > \gamma_{2m}$ for $a > 1$, and (2.7) implies that $\gamma_{2m} > \gamma_{2m-1}$ for m odd and $a > 1$. However, using w_j defined in Remark 2.4, for m odd and $j \geq 2$ it is $\gamma_{2^j m} > \gamma_{2^j m-1}$ only for $a > w_j$, and $\gamma_{2^j m} < \gamma_{2^j m-1}$ for $1 < a < w_j$, cf. Figure 1 and (1.15).

In Figure 1 there are shown the first intervals F_{kn} in the case $a = \frac{5}{4}$. The numbers over F_{kn} are the numbers k in dyadic representation written with n digits. The bold subintervals $F_{2k+1,n+1} \setminus F_{2k,n+1}$ are connected with the basic inequalities (2.2) since, in the case $1 < a < 2$, the statement in (4.7) for t is equivalent to (2.5) and therefore also to (2.2). The nested intervals (4.5) are connected with special series.

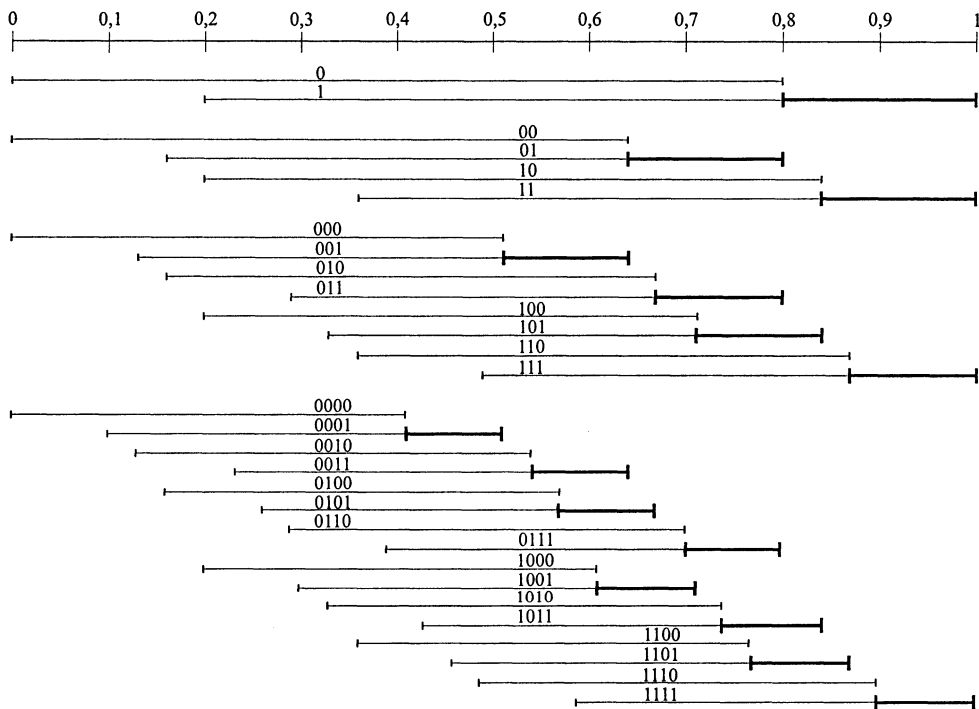


Figure 1: The intervals F_{kn} for $0 \leq k < 2^n$ and $1 \leq n \leq 4$

Proposition 4.4 *Let $a > 1$. The following assertions are equivalent:*

- (i) $t \in CM$,
- (ii) t is representable in the form $t = \{\mathring{F}_{k_n, n}\}$ with (4.5),
- (iii) t is representable in the form $t = \gamma_1 \sum_{\nu=1}^{\infty} \frac{\xi_\nu}{a^\nu}$, $\xi_\nu \in \{0, 1\}$ and ξ_ν not eventually constant.

The connection between (ii) and (iii) is given by

$$\frac{\gamma_{k_n}}{a^n} = \gamma_1 \sum_{\nu=1}^n \frac{\xi_\nu}{a^\nu} \quad (n \in \mathbb{N}_0). \quad (4.9)$$

Proof. (i) $\Rightarrow$ (ii): Every $t \in CM$ lies in $(0, 1) = \mathring{F}_{00}$. If $t \in \mathring{F}_{k_n}$ where $k = k_n$, then in view of (4.2) and (4.3) it must be $t \in \mathring{F}_{2k, n+1}$ or $t \in \mathring{F}_{2k+1, n+1}$, i.e. $t \in \mathring{F}_{k_{n+1}, n+1}$ for a certain k_{n+1} satisfying the condition in (4.5). Hence it holds (ii).

(ii) $\Rightarrow$ (iii): First we show that there exist numbers $\xi_\nu \in \{0, 1\}$ such that the numbers k_n in $F_{k_n, n}$ can be written as

$$k_n = \sum_{\nu=1}^n \xi_\nu 2^{n-\nu}, \quad (4.10)$$

i.e. on account of (1.12) such that

$$\gamma_{k_n} = \gamma_1 \sum_{\nu=0}^{n-1} \xi_{n-\nu} a^\nu. \quad (4.11)$$

Since this is trivial for $k_0 = 0$, we assume that (4.11) is valid for a fixed $n \in \mathbb{N}_0$. Then in view of (1.13) we have

$$\gamma_{2k_n} = a\gamma_{k_n} = \gamma_1 \sum_{\nu=0}^{n-1} \xi_{n-\nu} a^{\nu+1} = \gamma_1 \sum_{\nu=1}^n \xi_{n+1-\nu} a^\nu$$

and $\gamma_{2k_{n+1}} = \gamma_{2k_n} + \gamma_1$, i.e. we have (4.11) with $n+1$ instead of n , where $\xi_{n+1} = 0$ for $k_{n+1} = 2k_n$ and $\xi_{n+1} = 1$ for $k_{n+1} = 2k_n + 1$. Hence, (4.11) is proved by induction. The equations (4.11) imply (4.9) so that, by means of (4.1) and (ii), the representation of t in (iii) follows as $n \rightarrow \infty$. In the case $\xi_\nu = 0$ for $\nu > n$ we would have $t = \frac{\gamma_{k_n}}{a^n}$. In the case $\xi_\nu = 1$ for $\nu > n$ we would have $t = \frac{\gamma_{k_n+1}}{a^n}$ in view of

$$\gamma_1 \sum_{\nu=n+1}^{\infty} \frac{1}{a^\nu} = \frac{1}{a^n}. \quad (4.12)$$

However, both cases are contradictions to $t \in \overset{\circ}{F}_{k_n, n}$ so that they are impossible.

(iii) $\Rightarrow$ (i): From (iii), (4.1), (4.9) and (4.12) it follows $t \in \overset{\circ}{F}_{k_n, n}$ for all n . Hence (4.2) proves (i) ■

Remark 4.5 For the representation (iii) of t with the partial sums (4.9) we can say that $\xi_{n+1} = 0$ in the case (4.6) and $\xi_{n+1} = 1$ in the case (4.7). Hence, all coefficients are uniquely determined for $a \geq 2$, whereas ξ_{n+1} is not fixed for t contained in (4.4) and $1 < a < 2$. However, also in the last case it is possible to emphasize a unique representation (iii) for every $t \in (0, 1)$ by $\xi_{n+1} = 0$ for t contained in (4.4) with $k = k_n$, i.e. by

$$\xi_{n+1} = \begin{cases} 0 & \text{for } t \in \overset{\circ}{F}_{2k, n+1}, \\ 1 & \text{for } t \in \overset{\circ}{F}_{2k+1, n+1} \setminus \overset{\circ}{F}_{2k, n+1} \end{cases}$$

and all $n \in \mathbb{N}_0$. This choice of ξ_n yields equivalently for every $t \in (0, 1)$ a unique representation (ii) with $k_{n+1} = 2k_n$ for t contained in (4.4). Writing here m instead of n then the maximal possible j such that $k_{m+j} = 2^j k_m$ is the same as in Lemma 2.3 for a given $\ell = k_m$.

In the case $a \geq 2$ there do not exist representations (ii) for $t \in M$, but for every such t , i.e. for $t \in \overline{G}_{\ell m}$ with fixed ℓ, m , there exists a uniquely determined finite sequence $k_0, k_1, \dots, k_m = \ell$ with

$$\overset{\circ}{F}_{k_0, 0} \supset \overset{\circ}{F}_{k_1, 1} \supset \dots \supset \overset{\circ}{F}_{k_m, m} \supset \overline{G}_{\ell m}. \quad (4.13)$$

Writing $t = \frac{\gamma_{\ell} + \tau}{a^{m+1}}$ for $t \in \overline{G}_{\ell m}$ with $\ell = k_m$ and considering (4.1) and (4.9), we get the representation

$$t = \gamma_1 \sum_{\nu=1}^m \frac{\xi_\nu}{a^\nu} + \frac{\tau}{a^{m+1}} \quad (1 \leq \tau \leq \gamma_1), \quad (4.14)$$

cf. also [3], p.1007.

5. Modified expansions in the case $a \geq 2$

Lemma 4.2 has the following counterpart.

Lemma 5.1 *Let $a \geq 2$ and let k_n be the numbers appearing either in (ii) of Proposition 4.4 or in (4.13). In the latter case define $k_{m+1} = 2\ell + 1$. Then the solutions (k, n) of (2.2) are exactly the pairs (k_n, n) where k_n is followed by $k_{n+1} = 2k_n + 1$.*

Proof. The statement

$$t \in \overline{G}_{k_n} \cup \overset{\circ}{F}_{2k_n+1, n+1} = \left[\frac{a\gamma_k + 1}{a^{n+1}}, \frac{\gamma_k + 1}{a^n} \right) \quad (5.1)$$

is equivalent to (2.5) and therefore also to (2.2). According to (4.3) the statement (5.1) implies $t \in \overset{\circ}{F}_{k_n}$, and this proves the assertion ■

Remark 5.2 After defining $\xi_{m+1} = 1$ in the case (4.13) and using the connection (4.11), Remark 4.5 shows that the numbers k_n of Lemma 5.1 can also be characterized by $\xi_{n+1} = 1$.

After these preliminaries we can express the terms of the expansion (3.1) by means of the digits ξ_n . In the case (iii) we use the abbreviation $t_n = a^{n+1}t - a\gamma_{k_n}$ ($n \in \mathbb{N}_0$) with (4.11), i.e.

$$t_n = \gamma_1 \sum_{\nu=0}^{\infty} \frac{\xi_{n+\nu+1}}{a^\nu}. \quad (5.2)$$

Theorem 5.3 *Let $a \geq 2$. Then for $t \in CM$ with (iii) the expansion (3.1) can be modified to the alternating series*

$$\phi(t) = \sum_{n=0}^{\infty} (-1)^{\xi_1 + \dots + \xi_n} \xi_{n+1} f_n(t_n) \quad (5.3)$$

which satisfies the Leibniz condition for $n \geq 1$. For $t \in M$, i.e. $t \in \overline{G}_{\ell m}$ with (4.14), putting $\xi_{m+1} = 1$, expansion (3.1) terminates

$$\phi(t) = \sum_{n=0}^m (-1)^{\xi_1 + \dots + \xi_n} \xi_{n+1} f_n \left(\gamma_1 \sum_{\nu=0}^{m-n-1} \frac{\xi_{n+\nu+1}}{a^\nu} + \frac{\tau}{a^{m-n}} \right). \quad (5.4)$$

Proof. According to Lemma 2.5 we can write (3.1) as a series with respect to n . Lemma 5.1 yields $k = k_n$, Definition 2.2 and Remark 5.2 yield $\delta_{k_n} = \xi_{n+1}$, and by means of the definition of ε_k and (4.10) we obtain $\varepsilon_{k_n} = (-1)^{\xi_1 + \dots + \xi_n}$. Hence, (3.1) turns over into the equality

$$\phi(t) = \sum_{n=0}^m (-1)^{\xi_1 + \dots + \xi_n} \xi_{n+1} f_n(a^{n+1}t - \gamma_{2k_n}) \quad (5.5)$$

which is valid both for $t \in \overline{G}_{\ell m}$ and also for $t \in CM$ with $m = \infty$. From (5.5) it follows (5.3) for $t \in CM$ with (iii), and (5.4) for $t \in \overline{G}_{\ell m}$.

Obviously, (5.3) is an alternating series, if we regard only the non-vanishing terms. In order to show that it satisfies the Leibniz condition for $n \geq 1$ we assume that $\xi_{n+1} = 1$, $\xi_{n+m+1} = 1$ and $\xi_{n+\nu} = 0$ for $\nu = 2, \dots, m$ ($m \in \mathbb{N}$). Then the numbers (5.2) satisfy the inequalities

$$\gamma_1 \left(1 + \frac{1}{a^m}\right) \leq t_n < \gamma_1 \left(1 + \sum_{\nu=m}^{\infty} \frac{1}{a^\nu}\right) = \gamma_1 + \frac{1}{a^{m-1}},$$

and $t_{n+m} = a^m(t_n - \gamma_1)$. Applying Lemma 2.6 we have $f_n(t_n) > f_{n+m}(t_{n+m})$ for $n, m \in \mathbb{N}$, i.e. the series (5.3) satisfies indeed the Leibniz condition for $n \geq 1$ ■

Remark 5.4 1. In view of the Leibniz condition and (3.9) we can sharpen the estimate (3.8) in the case $a \geq 2$ by

$$0 < \varepsilon_{k_n} R_n < f_n(a^{n+1}t - \gamma_{2k_n}) < \frac{b^{n+1}}{n! a^{\frac{n(n-1)}{2}}}$$

where n must be chosen in such a way that k_n is defined.

2. The Leibniz condition is also fulfilled in the excluded case $(m, n) = (1, 0)$ of Lemma 2.6 if $f_0(t_0) \geq f_1(t_1)$ for $\xi_1 = \xi_2 = 1$ with t_n from (5.2), i.e. if

$$f_0(at) \geq f_1(a^2t - a^2 + a) \quad (5.6)$$

for $1 > t \geq \gamma_1(\frac{1}{a} + \frac{1}{a^2}) = 1 - \frac{1}{a^2}$. In view of $f_0 = b$ and $f_1(t) = \frac{b}{a-1}(t - \frac{1}{2})$ the inequality (5.6) means $a - 1 \geq a^2t - a^2 + a - \frac{1}{2}$, i.e. $t \leq 1 - \frac{1}{2a^2}$. Hence according to $a \geq 2$, the Leibniz condition for the series (5.3) is even satisfied for all terms so far as $t \leq \frac{7}{8}$. In view of the symmetry $\phi(t) = \phi(1 - t)$ this restriction is unessential.

3. The expansion (5.3) is valid not only for $t \in CM$ but also for the endpoints of $\overline{G}_{k_n}$, i.e. also for such t where we have eventually $\xi_n = 0$ or $\xi_n = 1$ in (iii), but then the finite representation (5.4) is to prefer.

4. Using (2.11) the expansion (5.3) can be transformed into

$$\phi(t) = \sum_{n=1}^{\infty} (-1)^{\xi_1 + \dots + \xi_n} (\xi_{n+1} - \xi_n) f_{n-1} \left(\frac{1}{a} t_n \right). \quad (5.7)$$

5. With other notations, representation (5.4) is already contained in [3] as (6.3).

6. The expansions (3.15) can analogously be modified as (3.1) in Theorem 5.3 and in (5.7).

Example 5.5 For the number

$$t = \frac{a-1}{a^{k-\ell}(a^\ell-1)} = \gamma_1 \sum_{n=0}^{\infty} \frac{1}{a^{k+\ell n}} \quad (k, \ell \in \mathbb{N})$$

we have that $\xi_{k+\ell n} = 1$ for $n \in \mathbb{N}_0$, whereas the remaining ξ_n in (iii) are equal to zero. Hence, for $a \geq 2$, the expansion (5.3) specializes to

$$\phi \left(\frac{a-1}{a^{k-\ell}(a^\ell-1)} \right) = \sum_{n=0}^{\infty} (-1)^n f_{k+\ell n-1} \left(\frac{a^\ell(a-1)}{a^\ell-1} \right). \quad (5.8)$$

In particular for $k = \ell = 2$ we obtain

$$\phi\left(\frac{1}{a+1}\right) = \sum_{n=0}^{\infty} (-1)^n f_{2n+1}\left(\frac{a^2}{a+1}\right). \quad (5.9)$$

For $\ell = 1$ and $k + 1$ instead of k we get, using $\phi(\frac{1}{a^k}) = f_{k-1}(1)$ from (2.10),

$$f_{k-1}(1) = \sum_{n=0}^{\infty} (-1)^n f_{k+n}(a).$$

This expansion also follows from (2.11) with $t = a$, therefore it is even valid for arbitrary $a > 1$ and for $k = 1$.

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